Linux Shell Cheat-Sheet: Beginner Edition

Bash essentials for daily use, interviews and on-call

The shell is the program that reads what you type and runs it. bash is the default on most Linux distros, so this sheet uses bash. `sh` / `dash` are smaller POSIX shells (used for scripts on Debian/Ubuntu), `zsh` is the macOS default, and `fish` is friendlier but not script-compatible. Everything below works in bash and zsh unless noted. Check yours with `echo $SHELL`.

Anatomy of a command: `command -options arguments`, for example `ls -l /var/log`. Add `--help` or run `man command` (quit with `q`) when you forget something.

1. Navigate

Command	What it does	Command	What it does
`pwd`	Show current directory	`cd ..`	Up one level
`ls`	List files	`cd ~ or cd`	Go to home directory
`ls -lah`	Long list, hidden files, readable sizes	`cd -`	Back to previous directory
`cd /var/log`	Go to a directory	`tree -L 2`	Show folder tree (if installed)

2. Files and Directories

Command	What it does	Command	What it does
`mkdir -p a/b/c`	Create nested directories	`rm file`	Delete file (no recycle bin)
`touch file`	Create empty file / update time	`rm -r dir`	Delete directory and contents
`cp src dst`	Copy file	`ln -s target link`	Create a symbolic link
`cp -r dir1 dir2`	Copy directory	`file name`	Show what type a file is
`mv old new`	Move or rename	`stat name`	Size, owner, permissions, times

Deleting is permanent. Never run `rm -rf` with a variable or `/` in it unless you have read it twice. Use `ls` with the same pattern first to preview what will match. Prefer `rm -i` while learning.

3. View and Search Files

Command	What it does	Command	What it does
`cat file`	Print whole file	`grep "text" file`	Lines containing text
`less file`	Scroll (q quit, / search)	`grep -rin "text" .`	Search folder, case-insens., line no.
`head -n 20 file`	First 20 lines	`grep -v "text"`	Lines NOT containing text
`tail -n 20 file`	Last 20 lines	`find . -name "*.log"`	Find files by name
`tail -f file`	Follow a log live (Ctrl-c stops)	`find / -size +100M`	Find files over 100 MB
`wc -l file`	Count lines	`which cmd`	Where a command lives

4. Pipes and Redirection

Command	What it does	Command	What it does
`cmd1 | cmd2`	Send output of cmd1 into cmd2	`cmd1 && cmd2`	Run cmd2 only if cmd1 succeeded
`cmd > file`	Write output to file (overwrites)	`cmd1 || cmd2`	Run cmd2 only if cmd1 failed

Command	What it does	Command	What it does
`cmd >> file`	Append output to file	`cmd1 ; cmd2`	Run both, one after the other
`cmd 2> err.txt`	Send errors to file	`cmd &`	Run in the background
`cmd > out 2>&1`	Send output and errors to one file	`sort | uniq -c`	Count duplicate lines
`cmd < file`	Read input from file	`cmd | tee file`	Show output and save it too

> Handy pipelines: `ps aux | grep nginx`, `ls -l | wc -l`, `cat access.log | sort | uniq -c | sort -rn | head`, `history | grep ssh`.

5. Permissions and Users

Command	What it does	Command	What it does
`ls -l`	Show permissions (rwxr-xr-x)	`chown user:group file`	Change owner and group
`chmod +x script.sh`	Make file executable	`sudo cmd`	Run one command as root
`chmod 644 file`	Owner rw, others read-only	`whoami`	Current user
`chmod 755 file`	Owner rwx, others rx	`id`	User ID and groups

> Permission numbers: read = 4, write = 2, execute = 1, added per group. Order is owner, group, others. So `754` = owner `rwx` (7), group `r-x` (5), others `r--` (4).

6. Processes and System

Command	What it does	Command	What it does
`ps aux`	List all processes	`df -h`	Disk space per filesystem
`top or htop`	Live process / CPU / memory view	`du -sh *`	Size of each item here
`kill PID`	Ask process to stop (SIGTERM)	`uptime`	Load and time since boot
`kill -9 PID`	Force kill (last resort)	`uname -a`	Kernel and OS info
`pkill name`	Kill by process name	`systemctl status svc`	Is a service running?
`jobs / fg / bg`	List / resume / background jobs	`sudo systemctl restart svc`	Restart a service
`free -h`	Memory usage	`journalctl -u svc -f`	Follow service logs

7. Network Basics

Command	What it does	Command	What it does
`ip a`	Show IP addresses	`ss -tulpn`	Listening ports and owners
`ping -c 4 host`	Test reachability	`ssh user@host`	Log in to remote machine
`curl -I url`	Fetch HTTP headers	`scp file user@host:/tmp/`	Copy file to remote host
`curl -O url`	Download a file	`dig example.com`	DNS lookup

8. Keyboard Shortcuts That Save Hours

Command	What it does	Command	What it does
`Tab`	Auto-complete (press twice for options)	`Ctrl-w`	Delete previous word
`Up / Down`	Previous / next command	`Ctrl-u`	Delete to start of line
`Ctrl-r`	Search command history	`Ctrl-z`	Pause job (resume with fg)
`Ctrl-c`	Stop the running command	`!!`	Repeat last command

Command	What it does	Command	What it does
`Ctrl-d`	Exit shell / end input	`sudo !!`	Repeat last command with sudo
`Ctrl-l`	Clear screen	`history`	List past commands
`Ctrl-a / Ctrl-e`	Jump to start / end of line	`!$`	Last argument of previous command

9. Variables, Aliases, Environment

Command	What it does	Command	What it does
`name="Ravi"`	Set variable (no spaces around =)	`export PATH=$PATH:~/bin`	Add a directory to PATH
`echo "$name"`	Print variable (quote it)	`alias ll="ls -lah"`	Create a shortcut
`export VAR=value`	Pass variable to child programs	`source ~/.bashrc`	Reload shell config
`env / printenv`	List environment variables	`$(command)`	Use command output inline
`echo $PATH`	Directories searched for commands	`echo $?`	Exit status of last command (0 = OK)

Config files: `~/.bashrc` (interactive shells, put aliases here), `~/.bash_profile` or `~/.profile` (login shells), `~/.zshrc` for zsh. Quoting: double quotes expand variables, single quotes do not.

10. Your First Script

```
#!/bin/bash
set -euo pipefail          # stop on errors and unset variables

name="${1:-world}"          # first argument, default "world"
echo "Hello, $name"

if [ -f /etc/os-release ]; then
    echo "OS info found"
fi

for f in *.log; do
    echo "Found log: $f"
done
```

Command	What it does	Command	What it does
`chmod +x hello.sh`	Make the script executable	`[ -f f ] [ -d d ]`	Test: file exists / dir exists
`./hello.sh Asha`	Run it with an argument	`[ "$a" = "$b" ]`	String equal (spaces required)
`$1 $2 $@ $#`	Arguments: first, second, all, count	`[ $n -gt 5 ]`	Number greater than 5 (-lt, -eq)

11. Habits That Keep You Safe

- Press `Tab` to complete names. It prevents typos and shows you what exists.
- Run `ls` before `rm`, and `pwd` before anything destructive. Know where you are.
- Quote variables: `"$file"`, not `$file`. Unquoted names with spaces break commands.
- Read before running anything copied from the internet, especially with `sudo` or `curl ... | bash`.
- Use `man cmd` or `cmd --help` first, and `tldr cmd` if installed for quick examples.
- Practise in a scratch folder (`mkdir ~/sandbox && cd ~/sandbox`) where mistakes cost nothing.